Lab 1 — Output Demonstration: Post-Installation System Verification and Directory Listing

ratul@firefly: ~ — bash

```
ratul@firefly:~$ uname -a
Linux firefly 7.0.0-38-generic #38-Ubuntu SMP PREEMPT_DYNAMIC Fri Sep  4 09:10:14 UTC 2026 x86_64 GNU/Linux
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ ls
Desktop   Documents   Downloads   Music   Pictures   Public   Templates   Videos   snap
ratul@firefly:~$ _
```

Lab 2 — Session 1: Finding Location, Moving Folders, and Listing Files

ratul@firefly: ~ — bash

```
ratul@firefly:~$ whoami
ratul
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ ls
Desktop   Documents   Downloads   lab_demo   Music   Pictures   Videos
ratul@firefly:~$ cd lab_demo
ratul@firefly:~/lab_demo$ pwd
/home/ratul/lab_demo
ratul@firefly:~/lab_demo$ ls
backups   data.csv   notes.txt   report.pdf   scripts
ratul@firefly:~/lab_demo$ ls -a
.   ..   .lab_profile   .secret.key   backups   data.csv   notes.txt   report.pdf   scripts
ratul@firefly:~/lab_demo$ _
```

Lab 2 — Session 2: Inspecting File Details, Easy Sizes, and Returning Home

ratul@firefly: ~/lab_demo — bash

```
ratul@firefly:~/lab_demo$ ls -l
total 23568
drwxr-xr-x 2 ratul users     4096 Oct  8 10:15 backups
-rw-r--r- 1 ratul users  14680064 Oct  8 10:20 data.csv
-rw-r--r- 1 ratul users     1433 Oct  8 10:22 notes.txt
-rw-r--r- 1 ratul users   262144 Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users     4096 Oct  8 10:18 scripts
ratul@firefly:~/lab_demo$ ls -lh
total 24M
drwxr-xr-x 2 ratul users 4.0K Oct  8 10:15 backups
-rw-r--r- 1 ratul users  14M Oct  8 10:20 data.csv
-rw-r--r- 1 ratul users 1.4K Oct  8 10:22 notes.txt
-rw-r--r- 1 ratul users 256K Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users 4.0K Oct  8 10:18 scripts
ratul@firefly:~/lab_demo$ ls -la
total 23580
drwxr-xr-x 4 ratul users     4096 Oct  8 10:28 .
drwx--- 9 ratul users     4096 Oct  8 09:50 ..
-rw-r--r- 1 ratul users      342 Oct  8 10:12 .lab_profile
-rw---- 1 ratul users       64 Oct  8 10:14 .secret.key
drwxr-xr-x 2 ratul users     4096 Oct  8 10:15 backups
-rw-r--r- 1 ratul users  14680064 Oct  8 10:20 data.csv
-rw-r--r- 1 ratul users     1433 Oct  8 10:22 notes.txt
-rw-r--r- 1 ratul users   262144 Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users     4096 Oct  8 10:18 scripts
ratul@firefly:~/lab_demo$ cd ~
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ _
```

Lab 3 — Session 1: Creating Directories, Nested Trees, Empty Files, and Copying

ratul@firefly: ~/lab3_demo — bash

```
ratul@firefly:~/lab3_demo$ mkdir docs backup
ratul@firefly:~/lab3_demo$ mkdir -p proj/src/utils
ratul@firefly:~/lab3_demo$ touch docs/notes.txt proj/src/main.c
ratul@firefly:~/lab3_demo$ ls docs proj/src
docs:  notes.txt        proj/src:  main.c  utils
ratul@firefly:~/lab3_demo$ cp docs/notes.txt docs/notes_copy.txt
ratul@firefly:~/lab3_demo$ ls -l docs
total 0
-rw-r--r- 1 ratul users 0 Oct  8 14:10 notes.txt
-rw-r--r- 1 ratul users 0 Oct  8 14:12 notes_copy.txt
ratul@firefly:~/lab3_demo$ _
```

Lab 3 — Session 2: Moving, Renaming, and Removing Files and Directories

ratul@firefly: ~/lab3_demo — bash

```
ratul@firefly:~/lab3_demo$ mv docs/notes_copy.txt backup/
ratul@firefly:~/lab3_demo$ mv backup/notes_copy.txt backup/notes_v1.txt
ratul@firefly:~/lab3_demo$ ls backup
notes_v1.txt
ratul@firefly:~/lab3_demo$ touch temp.log && rm temp.log
ratul@firefly:~/lab3_demo$ mkdir -p temp_cache/data && touch temp_cache/data/cache.bin
ratul@firefly:~/lab3_demo$ rm -r temp_cache
ratul@firefly:~/lab3_demo$ mkdir -p scratch/a/b && touch scratch/a/test.tmp
ratul@firefly:~/lab3_demo$ rm -rf scratch
ratul@firefly:~/lab3_demo$ ls
backup   docs   proj
ratul@firefly:~/lab3_demo$ _
```

Lab 4 — Part 1: Viewing Files with cat, head, tail, and less

ratul@firefly: ~/lab4 — bash

```
ratul@firefly:~/lab4$ cat server.log
[2026-10-08 09:00:01] INFO System startup complete
[2026-10-08 09:05:12] DEBUG Loading database connection pool
[2026-10-08 09:12:44] WARN High CPU usage detected on core 2
[2026-10-08 09:20:10] ERROR Port 8080 timeout connecting to worker
[2026-10-08 09:30:00] INFO Scheduled backup process finished
ratul@firefly:~/lab4$ head -n 2 server.log
[2026-10-08 09:00:01] INFO System startup complete
[2026-10-08 09:05:12] DEBUG Loading database connection pool
ratul@firefly:~/lab4$ tail -n 2 server.log
[2026-10-08 09:20:10] ERROR Port 8080 timeout connecting to worker
[2026-10-08 09:30:00] INFO Scheduled backup process finished
ratul@firefly:~/lab4$ less server.log
server.log (lines 1-5/5) (Press q to quit, / to search)
```

Lab 4 — Part 2: Installing and Editing with GNU Nano

ratul@firefly: ~/lab4 — bash

```
ratul@firefly:~/lab4$ which nano
/usr/bin/nano
ratul@firefly:~/lab4$ sudo apt install nano
nano is already the newest version (7.2-1ubuntu1).
ratul@firefly:~/lab4$ nano app_config.txt
GNU nano 7.2               File: app_config.txt               Modified
server_mode = production
max_connections = 500
port = 9000
[ Replace "development" with "production" completed: 1 replacement ]
^G Get Help  ^O WriteOut  ^W Where Is  ^K Cut     M-U Undo     ^X Exit
^R Read File ^\ Replace   ^_ Go To Line ^U Paste   M-E Redo     M-6 Copy
ratul@firefly:~/lab4$ cat app_config.txt
server_mode = production | max_connections = 500 | port = 9000
```

Lab 4 — Part 3: Installing and Editing with Vim

ratul@firefly: ~/lab4 — bash

```
ratul@firefly:~/lab4$ which vim
/usr/bin/vim
ratul@firefly:~/lab4$ sudo apt install vim
vim is already the newest version (2:9.1.0-1ubuntu1).
ratul@firefly:~/lab4$ vim script.sh
~                    VIM - Vi IMproved 9.1
1 #!/usr/bin/env bash
2 echo "Starting backup service..."
3 echo "Status: service running successfully."
4 echo "Completed backup."
- INSERT -                                          4,1       All
:%s/task/service/g   2 substitutions on 2 lines
:wq                  "script.sh" 4L, 114B written
ratul@firefly:~/lab4$ cat script.sh
#!/usr/bin/env bash
echo "Starting backup service..."
echo "Status: service running successfully."
echo "Completed backup."
ratul@firefly:~/lab4$ _
```

Lab 5 — Session 1: Finding Files by Name, Extension, and Type with find

ratul@firefly: ~/lab5_search — bash

```
ratul@firefly:~/lab5_search$ find . -name "*.log"
./logs/error.log
./logs/server.log
ratul@firefly:~/lab5_search$ find . -iname "readme*"
./docs/README.md
ratul@firefly:~/lab5_search$ find . -type d
.
./docs
./src
./logs
ratul@firefly:~/lab5_search$ find . -type f -name "*.py"
./src/main.py
./src/utils.py
ratul@firefly:~/lab5_search$ _
```

Lab 5 — Session 2: Searching File Content with grep, grep -r, and Flags

ratul@firefly: ~/lab5_search — bash

```
ratul@firefly:~/lab5_search$ grep "ERROR" logs/server.log
[09:15:20] ERROR Database connection pool exhausted on port 5432
[09:22:45] ERROR Worker node 3 failed health check
ratul@firefly:~/lab5_search$ grep -i "warning" logs/server.log
[09:10:02] WARN  High memory threshold exceeded: 82%
[09:12:30] Warning: Slow query response time (420ms)
ratul@firefly:~/lab5_search$ grep -n "failed" logs/server.log
24:[09:22:45] ERROR Worker node 3 failed health check
ratul@firefly:~/lab5_search$ grep -v "INFO" logs/server.log | head -n 2
[09:10:02] WARN High memory threshold exceeded: 82%
[09:15:20] ERROR Database connection pool exhausted on port 5432
ratul@firefly:~/lab5_search$ grep -r "database" .
src/main.py:DB_HOST = "database.internal.net"
docs/notes.txt:Remember to run database migrations before deploy
ratul@firefly:~/lab5_search$ grep -rn "TODO" .
src/utils.py:14:# TODO: add token authentication
docs/notes.txt:3:- TODO: verify backup cron job
ratul@firefly:~/lab5_search$ _
```

Lab 6 — Session 1: Updating Package Lists, Upgrading, and Installing Software

ratul@firefly: ~ — bash

```
ratul@firefly:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
All packages are up to date.
ratul@firefly:~$ sudo apt upgrade -y
Reading package lists... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ratul@firefly:~$ sudo apt install -y htop tree
Reading package lists... Done
The following NEW packages will be installed:
  htop tree
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 188 kB of archives.
Unpacking htop (3.3.0-4build1) ...
Unpacking tree (2.1.1-1) ...
Setting up htop (3.3.0-4build1) ...
Setting up tree (2.1.1-1) ...
ratul@firefly:~$ _
```

Lab 6 — Session 2: Uninstalling Packages, Cleaning Leftovers, and Full Purging

ratul@firefly: ~ — bash

```
ratul@firefly:~$ which htop tree
/usr/bin/htop
/usr/bin/tree
ratul@firefly:~$ sudo apt remove -y tree
Reading package lists... Done
The following packages will be REMOVED:
  tree
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
Removing tree (2.1.1-1) ...
Processing triggers for man-db (2.12.0-4build1) ...
ratul@firefly:~$ sudo apt autoremove -y
Reading package lists... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ratul@firefly:~$ sudo apt purge -y htop
Reading package lists... Done
The following packages will be REMOVED:
  htop*
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
Removing htop (3.3.0-4build1) ...
Purging configuration files for htop (3.3.0-4build1) ...
ratul@firefly:~$ which htop
(no output -- command not found)
ratul@firefly:~$ _
```

Lab 7 — Session 1: Identity, System Time, Output Display, and Documentation

ratul@firefly: ~ — bash

```
ratul@firefly:~$ hostname && whoami
firefly
ratul
ratul@firefly:~$ who
ratul   pts/0      2026-10-08 09:15 (:0)
ratul@firefly:~$ date
Thu Oct 8 10:30:15 UTC 2026
ratul@firefly:~$ echo "Testing delay..." && sleep 2 && echo "Ready!"
Testing delay...
Ready!
ratul@firefly:~$ clear
[Screen cleared successfully]
ratul@firefly:~$ man echo | head -n 4
ECHO(1)                    User Commands                    ECHO(1)
NAME
    echo - display a line of text
ratul@firefly:~$ _
```

Lab 7 — Session 2: Command Shortcuts (Alias), Environment Variables, and History

ratul@firefly: ~ — bash

```
ratul@firefly:~$ alias ll="ls -lh"
ratul@firefly:~$ ll
total 8.0K
drwxr-xr-x 2 ratul users 4.0K Oct  8 09:30 Documents
-rw-r--r-- 1 ratul users 1.2K Oct  8 10:12 notes.txt
ratul@firefly:~$ printenv USER SHELL
ratul
/bin/bash
ratul@firefly:~$ env | grep -E "USER|HOSTNAME|HOME"
HOME=/home/ratul
HOSTNAME=firefly
USER=ratul
ratul@firefly:~$ history | tail -n 6
  42  hostname && whoami
  43  date
  44  echo "Testing delay..." && sleep 2
  45  alias ll="ls -lh"
  46  printenv USER SHELL
  47  history | tail -n 6
ratul@firefly:~$ _
```